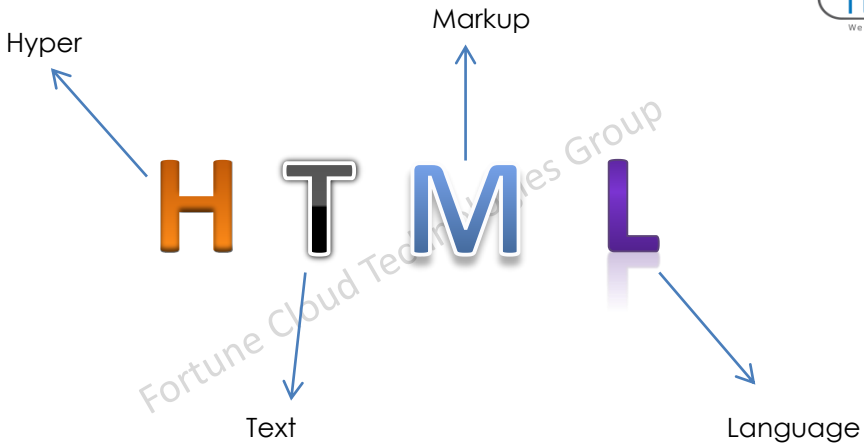


HTML Introduction



HTML

- HTML stands for HyperText Markup Language.
- HTML is used to create web pages and web applications.
- We can create a static website by HTML only.
- HTML is a Markup language rather than a programming language.

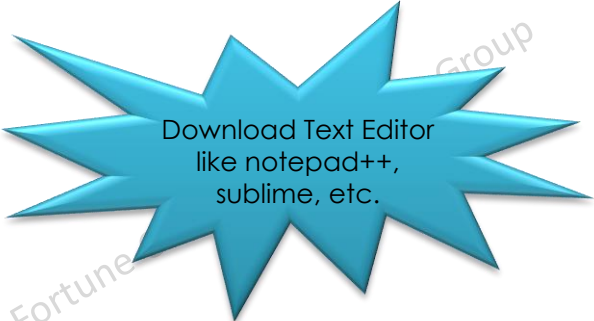
```
<!DOCTYPE>  
<html>  
  <head>  
    <title>Page title</title>  
  </head>  
  <body>  
  
    .....  
  
  </body>  
</html>
```



Demo.html

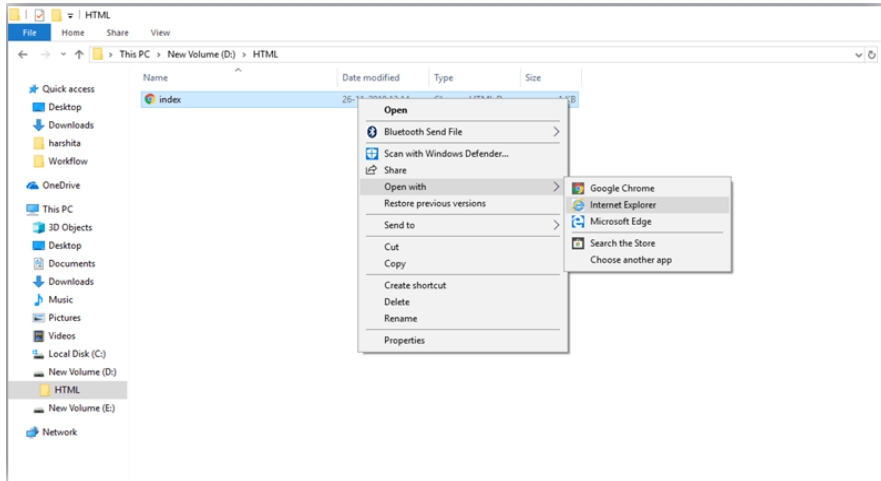
Features

- It is a very **easy and simple language**.
- It is very easy to make an **effective presentation**.
- It is a **markup language**, so it provides a flexible way to design web pages along with the text.
- It facilitates programmers to add a **link** on the web pages (by html anchor tag).
- It is **platform-independent**.
- It facilitates the programmer to add **Graphics, Videos, and Sound** to the web pages
- HTML is a case-insensitive language



Download Text Editor
like notepad++,
sublime, etc.

How to run



HTML Tags

- HTML tags are like keywords which defines that how web browser will format and display the content.



`<tag> content </tag>`

Unclosed HTML Tags

- **
 Tag:** br stands for break line, it breaks the line of the code.
- **<hr> Tag:** hr stands for Horizontal Rule. This tag is used to put a line across the webpage.



HTML Tags

HTML Heading

- A HTML heading or HTML h tag can be defined as a title or a subtitle
- There are six different HTML headings which are defined with the <h1> to <h6> tags

<h1>Heading no. 1</h1>

<h2>Heading no. 2</h2>

<h3>Heading no. 3</h3>

<h4>Heading no. 4</h4>

<h5>Heading no. 5</h5>

<h6>Heading no. 6</h6>

HTML Paragraph

- HTML paragraph or HTML p tag is used to define a paragraph in a webpage.
- An HTML `<p>` tag indicates starting of new paragraph.

```
<p>This is first paragraph.</p>
```

```
<p>This is second paragraph.</p>
```

Bold Text

- The HTML `<b>` element is a physical tag which display text in bold font

```
<b>bold text.</b>
```

- The HTML `<strong>` tag is a logical tag, which displays the content in bold font and informs the browser about its logical importance.

```
<strong>strong text.</strong>
```

Italic Text

- The HTML `<i>` element is physical element, which display the enclosed content in italic font

```
<i>Italic text.</i>
```

HTML Marked formatting

- If you want to mark or highlight a text, you should write the content within `<mark>.....</mark>`.

```
<h2> Hyper Text <mark> Markup</mark> Language</h2>
```

Underline Text

- If you write anything within `<u>.....</u>` element, is shown in underlined text.

`<u>` underlined text. `</u>`

- The **HTML <pre> tag** is used to specify pre formatted texts.
- Texts within <pre>.....</pre> tag is displayed in a fixed-width font.
- Usually it is displayed in Courier font.
- It maintains both space and line break.

```
<pre>
```

```
This is a formatted text
```

```
    This is a formatted text
```

```
        This is a formatted text
```

```
</pre>
```

HTML <sub> tag

- HTML <sub> tag is termed as Subscript tag and which is used to define subscript text.

H₂SO₄

HTML <sup> tag

- HTML <sup> tag is termed as a superscript tag which is used to define superscript text.

10th

12th

HTML Anchor

- The **HTML anchor tag** defines a *hyperlink that links one page to another page*.

```
<a href="second.html">Click for Second Page</a>
```

- If we want to open that link to another page then we can use target attribute of `<a>` tag.

```
<a href="second.html" target="_blank">Click for Second Page</a>
```

HTML Table

- **HTML table tag** is used to display data in tabular form (row * column). There can be many columns in a row.

```
<table>  
....  
</table>
```

```
<thead>  
  <tr>  
    <th>Id</th>  
    <th>Name</th>  
  </tr>  
</thead>
```

```
<tbody>  
  <tr>  
    <td>101</td>  
    <td>Abc</td>  
  </tr>  
</tbody>
```

```
<tfoot>  
  <tr>  
    <th>Id</th>  
    <th>Name</th>  
  </tr>  
</tfoot>
```

HTML Image

- **HTML img tag** is used to display image on the web page.
- Closing tags are not used in HTML image element.

```

```

- **src:** It is a necessary attribute that describes the source or path of the image.
- **alt:** The alt attribute defines an alternate text for the image, if it can't be displayed.

HTML Lists

Types of HTML Lists



Ordered List



Unordered List



Description
List or
Definition List

Ordered List

- In the ordered HTML lists, all the list items are marked with numbers by default.
- The ordered list starts with `<ol>` tag and the list items start with `<li>` tag.

```
<ol>  
  <li>Abc</li>  
  <li>Lmn</li>  
  <li>Pqr</li>  
  <li>Xyz</li>  
</ol>
```

```
type="1"
```

```
type="a"
```

```
type="A"
```

```
type="i"
```

Unordered List

- In HTML Unordered list, all the list items are marked with bullets.
- The Unordered list starts with `<ul>` tag and list items start with the `<li>` tag.

```
<ul>  
  <li>Abc</li>  
  <li>Lmn</li>  
  <li>Pqr</li>  
  <li>Xyz</li>  
</ul>
```

list-style-type:circle

list-style-type:disc

list-style-type:square

Description List or Definition List

- The definition list is very appropriate when you want to present glossary, list of terms or other name-value list.
- **<dl> tag** defines the start of the list.
- **<dt> tag** defines a term.
- **<dd> tag** defines the term definition (description).

```
<dl>
  <dt>Aries</dt>
  <dd>-One of the 12 horoscope sign.</dd>
  <dt>Bingo</dt>
  <dd>-One of my evening snacks</dd>
</dl>
```



HTML Form

- **HTML form** is a section of a document which contains controls such as text fields, password fields, checkboxes, radio buttons, submit button, menus etc.
- An HTML form facilitates the user to enter data that is to be sent to the server for processing
- HTML forms are required if you want to collect some data from of the site visitor.

```
<form action="" method="">  
    //input controls e.g. textfield, textarea, radiobutton, button  
</form>
```

HTML <input>

- It is used to create form fields, to take input from user.

```
<input type="text" name="name">
```

type="text"

type="email"

type="password"

type="number"

type="checkbox"

type="radio"

type="file"

HTML <textarea>

- <textarea> tag in HTML is used to insert multiple-line text in a form.
- The size of <textarea> can be specify either using "rows" or "cols" attribute

```
<textarea rows="7" cols="20"></textarea>
```



`<label> Tag`

- it makes the code parser/browser/user friendly.
- If you click on the label tag, it will focus on the text control.
- To do so, you need to have for attribute in label tag that must be same as id attribute of input tag.

```
<form>  
  <label for="firstname">First Name: </label> <br/>  
    <input type="text" id="firstname" name="firstname"/> <br/>  
  
  <label for="lastname">Last Name: </label>  
    <input type="text" id="lastname" name="lastname"/> <br/>  
</form>
```

<select> Tag

- HTML <select> tag is used to create a drop down list with multiple options.
- The <option> element is nested within <select> tag

```
<select>  
  <option>A</option>  
  <option>B</option>  
  <option>C</option>  
</select>
```

Submit Button

- it is used to add a submit button on web page.

```
<input type="submit" value="submit">
```

```
<button type="submit" >Submit</button>
```

HTML <fieldset>

- It is used to group the related information of a form.
- This element is used with <legend> element which provide caption for the grouped elements.

```
<form>  
  <fieldset>  
    <legend>User Information:</legend>  
    <label for="name">Enter name</label><br>  
    <input type="text" id="name" name="name"><br>  
  
    <input type="submit" value="submit">  
  </fieldset>  
</form>
```



HTML <abbr>

- HTML <abbr> tag is used to represent an acronym or abbreviation of a longer word or phrase, such as www, HTML, HTTP, etc.

<abbr title="HyperText Markup language">HTML</abbr>

HTML <ins>

- HTML <ins> tag is used to represent the newly added text in an HTML document.
- It usually renders with an underline through the text.

HTML

- HTML tag is used to represent the range of text that has been deleted/removed from the document.

If you will do **hard work <ins>smart work</ins>** then you will get success easily

- The **HTML <div> tag** is used to group the large section of HTML elements together.
- The div tag is generally used by web developers to group HTML elements together and apply CSS styles to many elements at once.

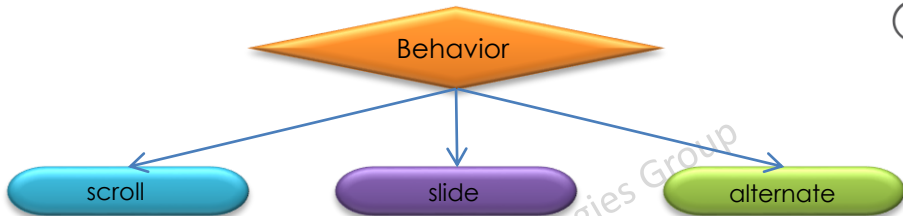
```
<div style="border:1px;padding:20px;font-size:20px">  
  <p>This is first paragraph</p>  
  <p>This is second paragraph</p>  
</div>
```



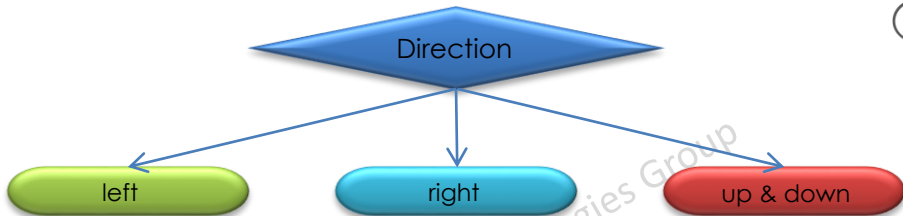
HTML <marquee>

- The **Marquee HTML** tag is a non-standard HTML element which is used to scroll a image or text horizontally or vertically.

<marquee>Hyper Text Markup Language**</marquee>**



```
<marquee width="100%" behavior="scroll" bgcolor="pink">  
  This is an example of a scroll marquee...  
</marquee>
```



<marquee width="100%" direction="right">
This is an example of a right direction marquee...
</marquee>

- HTML <nav> tag is used to represent a section which contains navigation links.

```
<nav>  
  <a href="#">Home</a> |  
  <a href="#">Courses</a> |  
  <a href="#">About-us</a> |  
  <a href="#">Contact-us</a> |  
</nav>
```

- **HTML audio tag** is used to define sounds such as music and other audio clips.
- Currently there are three supported file format for HTML 5 audio tag.
mp3, wav, ogg.

```
<audio controls>  
  <source src="birthday.mp3" type="audio/mp3">  
</audio>
```

Attributes

controls, autoplay, loop, muted, preload

HTML Video

- The HTML video tag is used for streaming video files such as a movie clip, song clip on the web page.
- Currently, there are three video formats supported for HTML video tag: Mp4, webM, ogg

```
<video controls>  
  <source src="movie.mp4" type="video/mp4">  
</video>
```

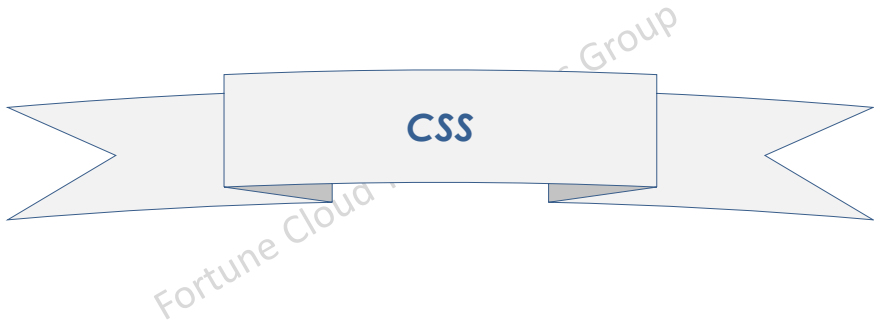
Attributes

controls, autoplay, loop, muted, preload

- HTML Iframe is used to display a nested webpage (a webpage within a webpage).
- An HTML iframe embeds another document within the current HTML document in the rectangular region.

```
<iframe src="sample.html" height="300" width="400"></iframe>
```

```
<iframe src="sample.html" height="50%" width="50%"></iframe>
```



- CSS stands for Cascading Style Sheet.
- CSS is used to design HTML tags.
- CSS is a widely used language on the web.
- HTML, CSS and JavaScript are used for web designing. It helps the web designers to apply style on HTML tags.

CSS Example

```
<style>
h1
{
  color:white;
  background-color:red;
}
</style>
```

```
<h1>Cascading Stylesheet</h1>
```

Ways to insert CSS



Inline CSS



Internal CSS



External CSS

Inline CSS

- Inline CSS is used to apply CSS on a single line or element.

```
<p style="color:red">Welcome</p>
```

```
<h1 style="color:blue">Welcome</h1>
```

Internal CSS

- It is written inside the style tag within head section of html.

```
<style>  
h1  
{  
  color:white;  
  background-color:red;  
}  
</style>
```

<h1>Cascading Stylesheet**</h1>**

External CSS

- Here, we write all the CSS code in a css file.
- Its extension must be .css

```
<link rel="stylesheet" type="text/css" href="style.css">
```

The link tag must be used inside head section of html.

style.css

```
P  
{  
    color:blue;  
}
```


CSS Selectors

Element
Selector

Id
Selector

Class
Selector

Universal
Selector

Group
Selector

Element Selector

- The element selector selects the HTML element by name.

```
<style>  
h1  
{  
  color:white;  
  background-color:red;  
}  
</style>
```

```
<h1>Cascading Stylesheet</h1>
```

Id Selector

- The id selector selects the id attribute of an HTML element to select a specific element.
- An id is always unique within the page

```
<style>  
#abc  
{  
  color:red;  
}  
</style>
```

```
<h1 id="abc">Cascading Stylesheet</h1>
```

```
<h1>Cascading Stylesheet</h1>
```

Class Selector

- The class selector selects HTML elements with a specific class attribute.
- It is used with a . (full stop symbol)

```
<style>  
.xyz  
{  
  color:red;  
}  
</style>
```

```
<h1 class="xyz">Cascading Stylesheet</h1>
```

```
<h1>Cascading Stylesheet</h1>
```

Universal Selector

- It selects all the elements on the pages.

```
<style>  
*  
{  
  color:red;  
}  
</style>
```

```
<h1>Cascading Stylesheet</h1>  
<p>This is heading</p>
```

Group Selector

- It select all the elements with the same style definitions
- Grouping selector is used to minimize the code.
- Commas are used to separate each selector in grouping.

```
<style>  
h1,h2,p  
{  
  color: blue;  
}  
</style>
```

```
<h1>Cascading Stylesheet</h1>  
  
<h2>Cascading Stylesheet</h2>  
  
<p>This is heading</p>
```



Properties

background-color

- The background-color property is used to specify the background color of the element.

```
<style>
```

```
p
```

```
{
```

```
  background-color: #b0d4de;
```

```
}
```

```
</style>
```

```
<p>This is an example of CSS background  
-color.</p>
```




- The background-image property is used to set an image as a background of an element.

```
<style>  
body {  
    background-image: url("test.jpg");  
}  
</style>
```

```
<body>  
  
</body>
```

CSS Border

- It is used to set the border on an element.

```
<style>  
p {  
  border-style: dotted;  
}  
</style>
```

<p>Cascading Stylesheet</p>

dotted

dashed

solid

double

inset

outset

none

border-radius

- This CSS property sets the rounded borders and provides the rounded corners around an element, tags, or div.

```
<style>  
div  
{  
  border-radius: 90px;  
  background: lightgreen;  
}  
</style>
```

```
<div>  
  <h2>CSS border-radius property</h2>  
</div>
```

border-top-left-radius:10px;

border-top-right-radius:10px;

border-bottom-right-radius:10px;

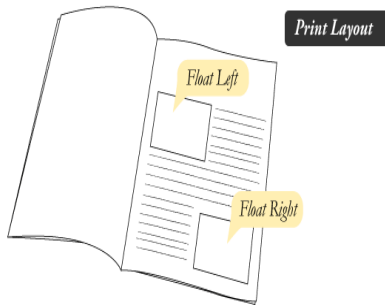
border-bottom-left-radius:10px;

- The **CSS float property** is a positioning property.
- It is used to push an element to the left or right, allowing other element to wrap around it.

```
<style>  
img {  
  float: right;  
}  
</style>
```

```

```



Font Color

- It is used to change the color of the text.
- There are three different formats to define a color;
 - By a color name
 - By hexadecimal value
 - By RGB

```
<style>  
  h1 { color: red; }  
  h2 { color: #9000A1; }  
  p { color:rgb(0, 220, 98); }  
</style>
```

```
<h1>This is heading 1</h1>  
  
<h2>This is heading 2</h2>  
  
<p>This is a paragraph.</p>
```

Font Family

```
<style>  
  h1 { font-family: sans-serif; }  
  h2 { font-family: serif; }  
  p { font-family: monospace; }  
</style>
```

```
<h1>This is heading 1</h1>  
  
<h2>This is heading 2</h2>  
  
<p>This is a paragraph.</p>
```

- CSS font size property is used to change the size of the font.

```
<style>  
p  
{  
  font-size:xx-small;  
}  
</style>
```

<p>This is a paragraph.**</p>**

xx-small

x-small

small

medium

large

x-large

xx-large

smaller

larger

size in pixels or %

Font Style

- CSS Font style property defines what type of font you want to display.
- It may be italic, oblique, or normal.

<style>

```
h1 { font-style: italic; }  
h2 { font-style: oblique; }  
p { font-style: normal; }
```

</style>

<h1>This is heading 1**</h1>**

<h2>This is heading 2**</h2>**

<p>This is a paragraph.**</p>**

CSS Hover

- The **:hover** selector is for selecting the elements when we move the mouse on them.
- the hover effect modifies the element's property value

```
<style>  
a:hover  
{  
  color: red;  
}  
</style>
```

```
<a href = "">Home</a>  
  
<a href = "">About</a>  
  
<a href = "">Contact</a>
```

CSS Important

- This property in CSS is used to give more importance compare to normal property.
- The ***!important*** means '**this is important**'.

```
<style>
h1
{
  font-size: 30px !important;
}
</style>
```

```
<h1>welcome</h1>
```

```
<h1>Hello</h1>
```

- CSS Margin property is used to define the space around elements.
- It is completely transparent and doesn't have any background color.

```
<style>
```

```
p
```

```
{
```

```
margin-top: 50px;
```

```
margin-bottom: 50px;
```

```
margin-right: 100px;
```

```
margin-left: 100px;
```

```
}
```

```
</style>
```

```
<p>welcome</p>
```

- **CSS Padding property** is used to define the space between the element content and the element border.

```
<style>
```

```
p
```

```
{
```

```
padding-top: 50px;
```

```
padding-right: 100px;
```

```
padding-bottom: 150px;
```

```
padding-left: 200px;
```

```
}
```

```
</style>
```

```
<p>welcome</p>
```

- The CSS opacity property is used to specify the transparency of an element.
- Opacity is defined as degree in which light is allowed to travel through an object.

```
<style>  
img  
{  
  opacity: 0.5;  
}  
</style>
```

```
>
```

Box - Shadow

- It is used to add shadow-like effects around the frame of an element.

```
<style>  
img  
{  
  box-shadow: 5px 10px 10px 10px orange;  
}  
</style>
```

```

```

Text - Shadow

- This CSS property adds shadows to the text.

```
<style>  
P  
{  
    text-shadow: 3px 3px red;  
}  
</style>
```

<p>Text Shadow </p>

- It is a graphical representation of a file or program that helps the user to identify about the type of file quickly.

```
<link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/4.7.0/css/font-awesome.min.css">
```

```
<i class="fa fa-envelope"></i>
```

```
<i class="fa fa-home"></i>
```

```
<i class="fa fa-phone"></i>
```


Letter-spacing

- This CSS property used to control the space between every letter inside an element or the block of text.
- Using this property, we can increase or decrease the space between the characters of the text.

```
<p style="letter-spacing: 7px;">This is a paragraph.</p>
```

Word-spacing

- This CSS property is used to control the space between the words.

```
<p style="word-spacing: 7px;">This is a paragraph.</p>
```

Navigation Bar

- A navigation bar is mostly displayed on the top of the page in the form of a horizontal list of links.

```
<style>
ul {
  list-style-type: none;
}

li {
  float: left;
}

li a {
  padding: 10px 20px;
  text-decoration: none;
}
</style>
```

```
<ul>
  <li><a href="#">Home</a></li>
  <li><a href="#">Java</a></li>
  <li><a href="#">HTML</a></li>
  <li><a href="#">CSS</a></li>
</ul>
```

Overlay

- it is used to set one thing on the top of another.
- Creating an overlay effect means to put two **div** together at the same place

```
<style>
img { width: 300px; height: 300px; }

.container { position: relative; width: 25%;
            height: auto; }

.overlay{
  position: absolute; transition: 0.5s ease;
  height: 300px; width: 300px; top: 0;
  background-color: blue; opacity: 0; }

.container:hover .overlay {
  opacity: 0.9; }
</style>
```

```
<div class="container">
  <img src= "test.jpg">
  <div class="overlay"></div>
</div>
```

Background-blend-mode

- We can blend the background images together.

```
<style>
#div1 {
  width: 400px;
  height: 400px;
  background-image: url("one.jpg"), url("two.jpg");
  background-blend-mode: multiply;
}
</style>
```

```
<div id="div1"></div>
```

Text-align

- It is used to set the horizontal alignment of a table-cell box or the block element.

```
<style>  
p {  
  text-align: center;  
}  
</style>
```

<p>Cascading Stylesheet</p>

center

left

right

justify

Columns

- It is used to set both **column-count** and **column-width** properties at the same time.
- The **column-count** property is used to set the number of columns.
- The **column-width** property specifies the width of the columns.

```
<style>
div {
  columns: 100px 4;
  border: solid 2px black;
  font-size: 20px;
}
</style>
```

```
<div>Your content...</div>
```

Animation

- An animation makes an element change gradually from one style to another.
- You can also specify the changes in percentage.
- 0% specify the start of the animation and 100% specify its completion.


```
<style>
div
{
  width: 150px;
  height: 150px;
  background: red;
  position: relative;
  -webkit-animation: myfirstanimation 5s;
}

@-webkit-keyframes myfirstanimation
{
  0% {background:red; left:0px; top:0px;}
  25% {background:yellow; left:300px; top:0px;}
  50% {background:blue; left:200px; top:300px;}
  75% {background:green; left:0px; top:200px;}
  100% {background:red; left:0px; top:0px;}
}

</style>
```



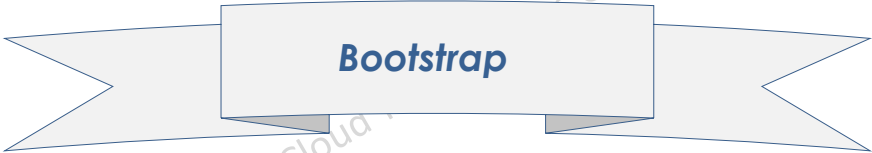
<div></div>

```
<style>
div
{
  width: 150px;
  height: 150px;
  background: red;
  position: relative;
  animation: myfirstanimation 5s;
}

@keyframes myfirstanimation
{
  0% {background:red; left:0px; top:0px;}
  25% {background:yellow; left:300px; top:0px;}
  50% {background:blue; left:300px; top:200px;}
  75% {background:green; left:0px; top:200px;}
  100% {background:red; left:0px; top:0px;}
}
</style>
```



`<div></div>`



Bootstrap

- Bootstrap is the popular HTML, CSS and JavaScript framework for developing a responsive and mobile friendly website
- It includes HTML and CSS based design templates for typography, forms, buttons, tables, navigation, modals, image carousels and many others.

What is a responsive website

- A website is called responsive website which can automatically adjust itself to look good on all devices, from smart phones to desktops etc.

Why use Bootstrap

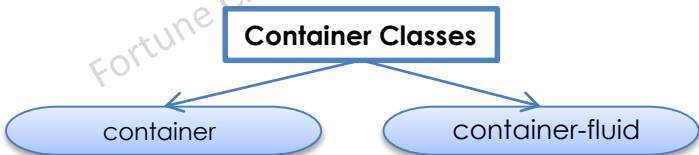
- It is very easy to use. Anybody having basic knowledge of HTML and CSS can use Bootstrap.
- It facilitates users to develop a responsive website.
- It is compatible on most of browsers like Chrome, Firefox, Internet Explorer, Safari and Opera etc.

Example

```
<html>
<head>
  <title>Bootstrap Example</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1">
  <link rel="stylesheet"
href="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/css/bootstrap.min.css">
  <script src="https://ajax.googleapis.com/ajax/libs/jquery/3.5.1/jquery.min.js"></script>
  <script src="https://maxcdn.bootstrapcdn.com/bootstrap/3.4.1/js/bootstrap.min.js"></script>
</head>
<body>
  <div class="container-fluid">
    <h1>Hello World!</h1>
  </div>
</body>
</html>
```

Bootstrap Container

- container is used to set the content's margins dealing with the responsive behaviors of your layout.
- It contains the row elements and the row elements are the container of columns (known as grid system).
- The **container class** is used to create boxed content.



Container-fluid

```
<div class="container-fluid">  
  <h1>Container-fluid</h1>  
  <p>container-fluid content</p>  
</div>
```

Container

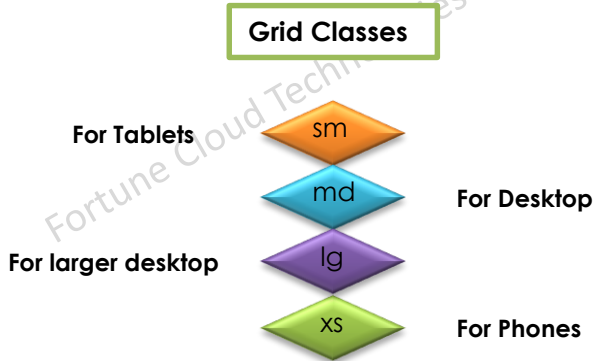
```
<div class="container">  
  <h1>Container</h1>  
  <p>container content</p>  
</div>
```


Grid System

- The Bootstrap Grid System allows up to 12 columns across the page.
- You can use all 12 columns individually or you can group the columns together

.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1	.col-md-1
.col-md-8								.col-md-4			
.col-md-4				.col-md-4				.col-md-4			
.col-md-6						.col-md-6					

- Bootstrap Grid System is responsive and the columns are re-arranged automatically according to the screen size.



Grid System

```
<div class="row">  
  <div class="col-sm-3">Division A</div>  
  <div class="col-sm-3">Division B</div>  
  <div class="col-sm-3">Division C</div>  
  <div class="col-sm-3">Division D</div>  
</div>
```

```
<div class="row">  
  <div class="col-sm-6">Division A</div>  
  <div class="col-sm-6">Division B</div>  
</div>
```

Button

```
<button type="button" class="btn btn-primary">Primary</button>
```

Primary

Secondary

Success

Danger

Warning

Info

Light

Dark

[Link](#)

Table

The **.table class** is used to add basic styling to a table.

```
<table class="table">  
  <tr>  
    <th>Id</th>  
    <th>Name</th>  
  </tr>  
  <tr>  
    <td>101</td>  
    <td>Abc</td>  
  </tr>  
  <tr>  
    <td>102</td>  
    <td>Xyz</td>  
  </tr>  
</table>
```

table table-striped

table table-bordered

table table-hover

Bootstrap Form Rules

- Always use `<form role="form">` (helps improve accessibility for people using screen readers)
- Wrap labels and form controls in `<div class="form-group">` (needed for optimum spacing)
- Add class `.form-control` to all textual `<input>`, `<textarea>`, and `<select>` elements

```
<div class="form-group">  
  <label for="inputEmail1">Email address</label>  
  <input type="email" class="form-control" id="inputEmail1" placeholder="Email">  
</div>  
  
<div class="form-group">  
  <label for="InputPassword">Password</label>  
  <input type="password" class="form-control" id="InputPassword" placeholder="Password">  
</div>  
  
<button type="submit" class="btn btn-default">Login</button>
```

Alert

- Bootstrap Alerts are used to provide an easy way to create predefined alert messages. Alert adds a style to your messages to make it more appealing to the users.
- There are four classes that are used within `<div>` element for alerts.

`.alert-success`

`.alert-info`

`.alert-warning`

`.alert-danger`

```
<div class="alert alert-success">  
  <strong>Success!</strong> Registration Successfully Completed  
</div>
```


Closing Alert

- If you want to close the alert message, you have to add an `.alert-dismissible` class to the alert container.

```
<div class="alert alert-success alert-dismissible ">  
  <button type="button" class="close" data-dismiss="alert">×</button>  
  <strong>Success!</strong> Registration Successfully Completed  
</div>
```

Panel

- A panel is a bordered box with some padding around its element.
- The panel components are used when you want to put your DOM component in a box.
- The **class .panel** is used within the <div> element to create Bootstrap panels. The content inside the panel has a **.panel-body class**.

Panel contains three parts

panel header

panel content

panel footer

Panel

```
<div class="panel panel-primary">  
  <div class="panel-heading">  
    <h3 class="panel-title">Panel title</h3>  
  </div>  
  
  <div class="panel-body">  
    Panel content  
  </div>  
  
  <div class="panel-footer">  
    Panel footer  
  </div>  
</div>
```

```
<div class="panel-group">
  <div class="panel panel-default">
    <div class="panel-heading">
      <h4 class="panel-title">
        <a data-toggle="collapse" href="#collapse1">Collapsible panel</a>
      </h4>
    </div>
    <div id="collapse1" class="panel-collapse collapse">
      <div class="panel-body">Panel Body</div>
      <div class="panel-footer">Panel Footer</div>
    </div>
  </div>
</div>
```

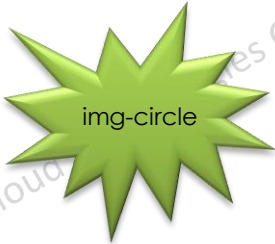
Image

Classes



img-rounded

- It adds border-radius
- image rounded corners



img-circle

- It makes the entire image round by adding border-radius



img-thumbnail

- It adds a bit of padding and a gray border.

Image

```

```

```

```

```

```

Glyphicons

- Glyphicons are the icon fonts that are used in web projects.

Syntax

```
<span class="glyphicon glyphicon-name"></span>
```

Note: To create the desired Glyphicon, the "name" part of the syntax must be replaced accordingly.

Examples

```
<span class="glyphicon glyphicon-envelope"></span>
```

```
<span class="glyphicon glyphicon-earphone"></span>
```

```
<span class="glyphicon glyphicon-log-in"></span>
```

```
<span class="glyphicon glyphicon-log-out"></span>
```

```
<span class="glyphicon glyphicon-save"></span>
```


Carousel slider

- The Bootstrap carousel is a flexible, responsive way that is used to add a slider to your webpage.
- It is very responsive and flexible enough to allow, images, iframes, videos, or any other type of content that you want to add.

```
<div id="myCarousel" class="carousel slide" data-ride="carousel">
  <div class="carousel-inner" role="listbox">
    <div class="item active">
      
    </div>
    <div class="item">
      
    </div>
  </div>

  <a class="left carousel-control" href="#myCarousel" role="button" data-slide="prev">
    <span class="glyphicon glyphicon-chevron-left" aria-hidden="true"></span>
    <span class="sr-only">Previous</span>
  </a>
  <a class="right carousel-control" href="#myCarousel" role="button" data-slide="next">
    <span class="glyphicon glyphicon-chevron-right" aria-hidden="true"></span>
    <span class="sr-only">Next</span>
  </a>
</div>
```

Progress Bar

- The progress bar shows how far a user is in a process.

Colored Progress Bar

progress-bar-success

progress-bar-info

progress-bar-danger

progress-bar-warning

```
<div class="progress">  
  <div class="progress-bar progress-bar-success" role="progressbar" aria-valuenow="50"  
    aria-valuemin="0" aria-valuemax="100" style="width:50%">  
    50% Complete (success)  
  </div>  
</div>
```

Animated Progress Bar

- Use class **.active** to create animated progress bar.

```
<div class="progress">  
  <div class="progress-bar progress-bar-striped active" role="progressbar"  
    aria-valuenow="70" aria-valuemin="0" aria-valuemax="100" style="width:70%">  
    70%  
  </div>  
</div>
```

```
<nav class="navbar navbar-inverse">
  <div class="container-fluid">
    <div class="navbar-header">
      <button type="button" class="navbar-toggle" data-
toggle="collapse" data-target="#myNavbar">
        <span class="icon-bar"></span>
        <span class="icon-bar"></span>
        <span class="icon-bar"></span>
      </button>
      <a class="navbar-brand" href="#">WebSiteName</a>
    </div>
```

```
<div class="collapse navbar-collapse" id="myNavbar">
  <ul class="nav navbar-nav">
    <li class="active"><a href="#">Home</a></li>
    <li class="dropdown">
      <a class="dropdown-toggle" data-toggle="dropdown" href="#">Categories
    <span class="caret"></span></a>
      <ul class="dropdown-menu">
        <li><a href="#">Page 1-1</a></li>
        <li><a href="#">Page 1-2</a></li>
      </ul>
    </li>
  </ul>
</div>
</div>
</nav>
```

Modal

- The bootstrap modal plugin is a dialog box / popup window that is displayed on top of the current page.

```
<button type="button" class="btn btn-info btn-lg" data-toggle="modal" data-target="#myModal">Open Modal</button>
```

```
<div class="modal fade" id="myModal" role="dialog">
  <div class="modal-dialog">
    <div class="modal-content">
      <div class="modal-header">
        <button type="button" class="close" data-dismiss="modal">×</button>
        <h4 class="modal-title">Modal Header</h4>
      </div>
      <div class="modal-body">
        <p>Write your text in the modal.</p>
      </div>
      <div class="modal-footer">
        <button type="button" class="btn btn-default" data-dismiss="modal">Close</button>
      </div>
    </div>
  </div>
</div>
```


JavaScript



- JavaScript (js) is a light-weight object-oriented programming language
- JavaScript is not a compiled language
- enables dynamic interactivity on websites when applied to an HTML document.



Features

- All popular web browsers support JavaScript
- JavaScript follows the syntax and structure of the C programming language
- JavaScript is an object-oriented programming language
- It is a case-sensitive language.



Applications

- Client-side validation
- Dynamic drop-down menus
- Displaying date and time
- Displaying pop-up windows and dialog boxes (like an alert dialog box, confirm dialog box and prompt dialog box)
- Displaying clocks etc.

How to write program in JavaScript

Demo.html

```
<script>  
    document.write("Hello JavaScript");  
    alert("Hello JavaScript ");  
</script>
```

External JavaScript

index.html

```
<html>
<head>
  <script type="text/javascript" src="ext.js"></script>
</head>
<body>
  <p>Welcome to JavaScript</p>
  <form>
    <input type="button" value="click" onclick="msg()"/>
  </form>
</body>
</html>
```

ext.js

```
function msg()
{
  alert("Hello JavaScript");
}
```

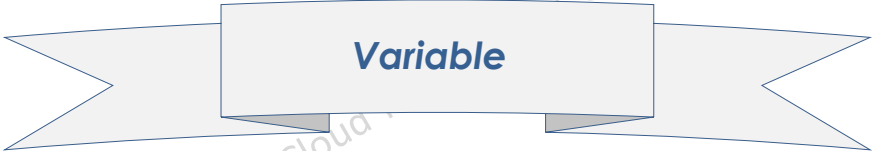
Comments

Single Line

```
<script>  
  // It is single line comment  
</script>
```

Multi Line

```
<script>  
  /*  
    It is multi line comment.  
    It will not be displayed.  
  */  
</script>
```



Variable

- A **variable** is simply a name of storage location.



Rules

- Name must start with a letter (a to z or A to Z), underscore (_), or dollar (\$) sign.
- After first letter we can use digits (0 to 9)
- JavaScript variables are case sensitive

```
<script>
```

```
var x = 10;
```

```
var y = 20;
```

```
var z=x+y;
```

```
document.write(z);
```

```
</script>
```



Local Variable

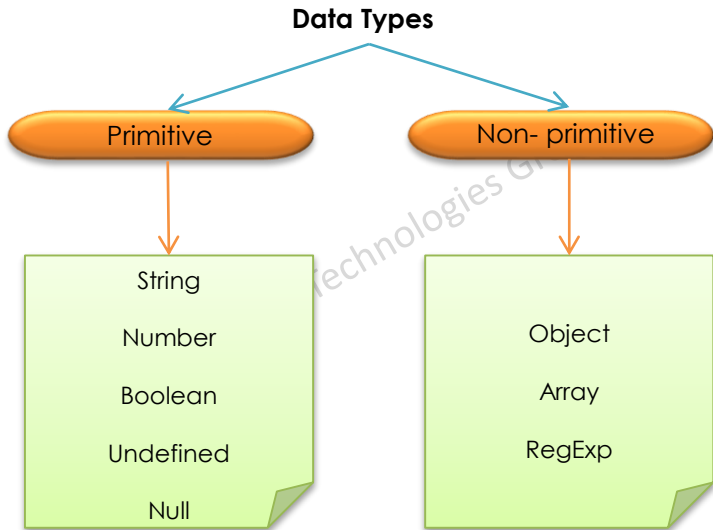
- Declared inside block or function
- Accessible within the function or block only

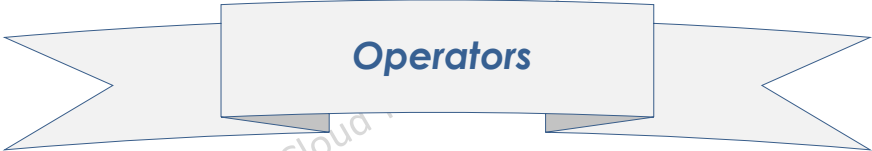
```
<script>  
  
function display()  
{  
    var a=10; //local variable  
}  
  
</script>
```

Global Variable

- Declared outside the function
- A **global variable** is accessible from any function

```
<script>  
  
var a=10; //Global variable  
  
function display()  
{  
    document.writeln(a);  
}  
</script>
```





Operators

Arithmetic

Operator	Description
+	Addition
-	Subtraction
*	Multiplication
/	Division
%	Modulus
++	Increment
--	Decrement

Comparison

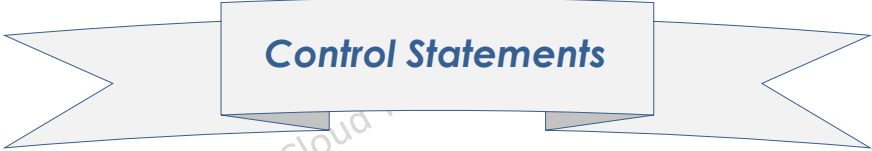
Operator	Description
==	Is equal to
===	Identical
!=	Not equal to
!==	Not Identical
>	Greater than
>=	Greater than or equal to
<	Less than

Bitwise

Operator	Description
&	Bitwise AND
	Bitwise OR
^	Bitwise XOR
~	Bitwise NOT
<<	Bitwise Left Shift
>>	Bitwise Right Shift

Logical

Operator	Description
&&	Logical AND
	Logical OR
!	Logical Not



Control Statements

if

```
if (expression)
{
    Statement(s) to be executed if expression is true
}
```

if...else

```
if (expression)
{
    Statement(s) to be executed if expression is true
}
else
{
    Statement(s) to be executed if expression is false
}
```

if...else if...

```
if (expression 1)
{
    Statement(s) to be executed if expression 1 is true
}
else if (expression 2)
{
    Statement(s) to be executed if expression 2 is true
}
else if (expression 3)
{
    Statement(s) to be executed if expression 3 is true
}
else
{
    Statement(s) to be executed if no expression is true
}
```

switch

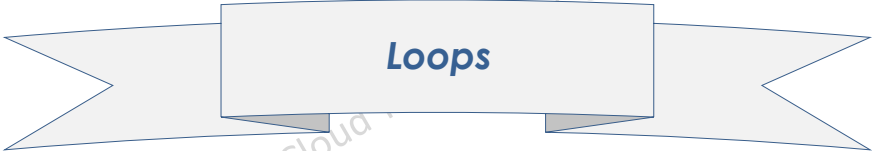
```
switch (expression)
{
    case condition 1:
        statement(s)
        break;

    case condition 2:
        statement(s)
        break;

    ...

    case condition n:
        statement(s)
        break;

    default:
        statement(s)
}
```



Loops

for

```
for (initialization; test condition; iteration statement)
{
    Statement(s) to be executed if test condition is true
}
```

while

```
while (expression)
{
    Statement(s) to be executed
}
```

do...while

```
do
{
    Statement(s) to be executed;
} while (expression);
```



Functions

- A function is a group of reusable code which can be called anywhere in your program.
- This eliminates the need of writing the same code again and again.
- Functions allow a programmer to divide a big program into a number of small and manageable functions.

Default Function

index.html

```
<html>
<head>
  <script type="text/javascript" src="ext.js"></script>
</head>
<body>
  <input type="button" value="click" onclick="show()"/>
</body>
</html>
```

ext.js

```
function show()
{
  alert("Default Function");
}
```

index.html

```
<html>
<head>
  <script type="text/javascript" src="ext.js"></script>
</head>
<body>
  <input type="button" value="click" onclick="addition(10,20)"/>
</body>
</html>
```

ext.js

```
function addition(a,b)
{
  var c=a+b;

  alert("Addition="+c);
}
```

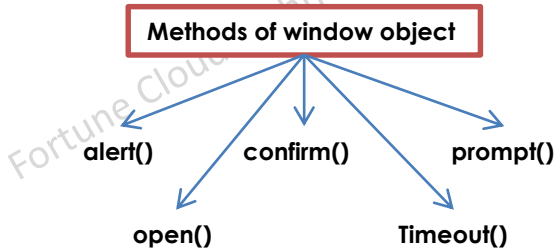
Return Statement

```
<input type="button" onclick="showme()" value="Return Function">

<script type = "text/javascript">
    function concatenate(first, last)
    {
        var fullname;
        fullname = first + last;
        return fullname;
    }

    function showme()
    {
        var result;
        result = concatenate('abc', 'def');
        document.write (result );
    }
</script>
```

- The **Browser Object Model** (BOM) is used to interact with the browser.
- The default object of browser is **window**.





alert()

```
<input type="button" onclick="clickme()" value="Alert">  
  
<script type = "text/javascript">  
  
    function clickme()  
    {  
        alert("I am alert");  
    }  
  
</script>
```



confirm()

```
<input type="button" onclick="clickme()" value="Confirm Alert">
```

```
<script type = "text/javascript">
```

```
    function clickme()  
    {  
        var v= confirm("Do you want to delete ?");  
        if(v==true)  
        {  
            alert("ok");  
        }  
        else  
        {  
            alert("cancel");  
        }  
    }
```

```
</script>
```



prompt()

```
<input type="button" onclick="clickme()" value="Prompt Alert">
```

```
<script type = "text/javascript">
```

```
    function clickme()
```

```
    {
```

```
        var name= prompt("What is your name?");
```

```
        alert("My Name is"+name);
```

```
    }
```

```
</script>
```



- It displays the content in a new window.

```
<input type="button" onclick="clickme()" value="Open">  
  
<script type = "text/javascript">  
  
    function clickme()  
    {  
        open("https://www.facebook.com/");  
    }  
</script>
```


Timeout()

- It performs its task after the given milliseconds.

```
<input type="button" onclick="clickme()" value="Open">  
  
<script type = "text/javascript">  
  
    function clickme()  
    {  
        setTimeout( function()  
        {  
            alert("I am Timeout");  
        },2000);  
    }  
</script>
```

DOM - Document Object Model

- The **document object** represents the whole html document.

```
<form name="myform">  
  <input type="text" name="firstname"/>  
  <button type="button" onclick="validation()" > DOM </button>  
</form>
```

```
<script type="text/javascript">  
  function validation()  
  {  
    var fname=document.myform.firstname.value;  
    alert("My name is: "+fname);  
  }  
</script>
```

Methods

`document.getElementById()`

`document.getElementsByName()`

`document.getElementsByTagName()`



`document.getElementById()`

- Return the element that has the ID attribute with the specified value.

```
<script type="text/javascript">  
function show()  
{  
    var num=document.getElementById("number").value;  
    alert(num);  
}  
</script>
```

```
<input type="text" id="number" />
```

```
<input type="button" value="submit" onclick="show()"/>
```



document.getElementsByName()

- Returns collection of all elements of particular document by name.

```
<script type="text/javascript">  
function show()  
{  
    var x=document.getElementsByName("abc");  
    alert("Total elements are:"+x.length);  
}  
</script>
```

```
<h4 name="abc">I</h4>  
<h4 name="abc">Am</h4>  
<h4 name="abc">Javascript</h4>
```

```
<input type="button" value="submit" onclick="show()"/>
```



`document.getElementsByTagName()`

- Returns the collection of all the elements in the document with the given tag name.

```
<script type="text/javascript">  
function show()  
{  
    var y=document.getElementsByTagName("h4");  
    alert("Total elements are:"+y.length);  
}  
</script>
```

```
<h4>I</h4>  
<h4>Am</h4>  
<h4>Javascript</h4>
```

```
<input type="button" value="submit" onclick="show()"/>
```

innerHTML

- This property is used to **set or return** the HTML content inside an element.

```
<script type="text/javascript">  
function show()  
{  
    var data="<input type='text' name='name'><br><br><input  
type='submit' value='Submit'>";  
    document.getElementById('showdiv').innerHTML=data;  
}  
</script>
```

```
<input type="button" value="submit" onclick="show()"/>  
  
<div id="showdiv"></div>
```



innerText

- Generate the dynamic content such as writing the validation message

```
<script type="text/javascript">
function validate()
{
    var name=document.getElementById('name').value;
    var msg="";
    if(name.length>3)
    {
        msg="correct";
    }
    else
    {
        msg="poor";
    }
    document.getElementById('error').innerText=msg;
}
</script>
```

```
<input type="text" id="name" onkeyup="validate()">

<span id="error"></span>
```




Jquery

- jQuery is a framework built using JavaScript capabilities.
- you can use all the functions and other capabilities available in JavaScript.
- jQuery is a small and lightweight JavaScript library.
- jQuery simplifies DOM manipulation.

Example

```
<html>
<head>
  <title>Home Page</title>
  <script type="text/javascript" src="http://ajax.googleapis.com/ajax/libs/jquery
/2.1.3/jquery.min.js"></script>

  <script type="text/javascript" language="javascript">
    $(document).ready(function() {
      $("p").css("background-color", "cyan");
    });
  </script>
</head>
<body>

  <p>The first paragraph is selected.</p>

</body>
</html>
```

Hide & Show

```
<script>  
$(document).ready(function(){  
  
    $("#hide").click(function(){  
        $("p").hide();  
    });  
  
    $("#show").click(function(){  
        $("p").show();  
    });  
  
});  
</script>
```

<p> Hello JQuery **</p>**

<button id="hide">Hide**</button>**
<button id="show">Show**</button>**

Effects

toggle

- It shows the hidden elements and hides the shown element.

```
<script>  
$(document).ready(function(){  
    $("button").click(function(){  
        $("p").toggle();  
    });  
});  
</script>
```

<p> Hello JQuery **</p>**

<button>Toggle**</button>**

slideDown

- jQuery slideDown() method is used to slide down an element.

```
<script>  
$(document).ready(function(){  
  
    $("#p1").click(function(){  
        $("#p2").slideDown("slow");  
    });  
  
});  
</script>
```

```
<style>  
#p2  
{  
    display: none;  
}  
</style>
```

<p id="p1"> Click Here </p>

<p id="p2"> Hello JQuery </p>

slideUp

- jQuery slideDown() method is used to slide up an element.

```
<script>
$(document).ready(function(){

    $("#p1").click(function(){
        $("#p2").slideUp("slow");
    });

});
</script>
```

<p id="p1"> Click Here </p>

<p id="p2"> Hello JQuery </p>

slideToggle

- If the element is slide down, it will slide up the element and if it is slide up, it will slide down.

```
<script>  
$(document).ready(function(){  
  
    $("#p1").click(function(){  
        $("#p2").slideToggle("slow");  
    });  
  
});  
</script>
```

```
<style>  
#p2  
{  
    display: none;  
}  
</style>
```

<p id="p1"> Click Here </p>

<p id="p2"> Hello JQuery </p>

Method

html

- It replaces the selected element content with new contents.

```
<script>  
$(document).ready(function(){  
  
    $("button").click(function(){  
        $("p").html("jQuery");  
    });  
  
});  
</script>
```

<button>Click here**</button>**

<p>This is a paragraph.**</p>**

Method

val

- It is used to set the value of every matched element.
- Get current value of the first element in the set of matched elements.

```
<script>  
function show()  
{  
  var sel = $( "#dropdown" ).val();  
  alert(sel);  
}  
  
$( "select" ).change(show);  
show();  
</script>
```

```
<select id="dropdown"  
  <option>Abc</option>  
  <option>Pqr</option>  
  <option>Lmn</option>  
  <option>Xyz</option>  
</select>
```

Method

append

- Insert specified content as the last child

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("ol").append("<li>New item</li>");
    });
});
</script>
```

```
<ol>
    <li>Abc</li>
    <li>Pqr</li>
    <li>Lmn</li>
    <li>Xyz</li>
</ol>

<button>Append</button>
```

Method

Clone

- It also makes copies of their child nodes

```
<script>  
$(document).ready(function(){  
    $("button").click(function(){  
        $("p").clone().appendTo("body");  
    });  
});  
</script>
```

<p>I am clone method**</p>**

<button>Clone**</button>**

Method

remove

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $( "p" ).remove();
    });
});
</script>
```

<p>I am remove method**</p>**

<button>Remove**</button>**

Method

empty

- Remove all content from the selected elements.

```
<script>
$(document).ready(function(){
    $("button").click(function(){
        $("div").empty();
    });
});
</script>
```

```
<div style="height:150px;background-color:red">
I am empty
</div>
```

```
<button>Empty</button>
```

Method

addClass

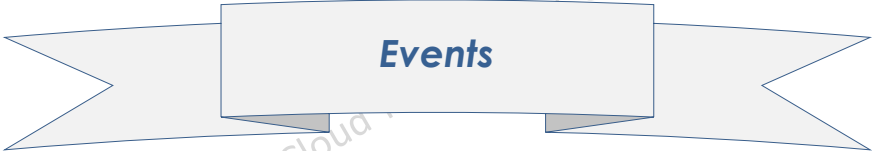
- The addClass() method is used to add one or more class name to the selected element.

```
<script>  
$(document).ready(function(){  
    $("button").click(function(){  
        $("p ").addClass("test");  
    });  
});  
</script>
```

```
<style>  
.test  
{  
    font-size: 50px;  
    color: red;  
}  
</style>
```

```
<p>This is a paragraph.</p>
```

```
<button>addClass</button>
```



Events

Event

click()

- When you click on an element, the click event occurs

```
<script>  
$(document).ready(function(){  
    $("button").click(function(){  
        alert("I am click event");  
    });  
});  
</script>
```

<button>Click here**</button>**

Event

focus()

- The jQuery focus event occurs when an element gains focus.

```
<input type="text">  
<span>What is your name ?</span>
```

```
<style>  
span  
{  
  display: none;  
}  
</style>
```

```
<script>  
$( "input" ).focus(function() {  
    $( this ).next( "span" ).css( "display", "inline" ).fadeOut( 2000 );  
});  
</script>
```

Event

change()

```
<script>  
function show()  
{  
  var sel = $( "#dropdown" ).val();  
  alert(sel);  
}  
  
$( "select" ).change(show);  
show();  
</script>
```

```
<select id="dropdown">  
  <option>Abc</option>  
  <option>Pqr</option>  
  <option>Lmn</option>  
  <option>Xyz</option>  
</select>
```

Event

keydown() & keyup()

```
<script>
$(document).ready(function(){
    $("input").keydown(function(){
        $("input").css("background-color", "green");
    });

    $("input").keyup(function(){
        $("input").css("background-color", "red");
    });
});
</script>
```

<input type="text">

Event

keypress()

- The jQuery keypress () event is occurred when a keyboard button is pressed down.

```
<script>  
i = 0;  
$(document).ready(function(){  
    $("input").keypress(function(){  
        $("span").text (i += 1);  
    });  
});  
</script>
```

<input type="text">

0

Event

mouseover() & mouseout()

- The mouseover event is occurred when you put your mouse cursor over the selected element .
- The mouseout event is occurred when you remove your mouse cursor from the selected element .

<script>

```
$(document).ready(function(){  
    $("p").mouseover(function(){  
        $("p").css("background-color", "lightgreen");  
    });  
    $("p").mouseout(function(){  
        $("p").css("background-color", "orange");  
    });  
});  
</script>
```

<p>This is a paragraph.**</p>**

Event

load()

- The load event occurs when a specific element is loaded.
- It is generally used with a URL (image, script, frame, iframe), and the window object.

```
<script>  
$(document).ready(function(){  
    $("img").load(function(){  
        alert("Image loaded.");  
    });  
});  
</script>
```

```

```



Thank you !